

Kartik Dangi

Robotics Engineer | M.Eng. Candidate, THWS

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SUMMARY

Robotics engineer (B.Eng. Robotics, THWS) spanning sensor fusion, ROS 2, and robot learning; admitted to the TUM M.Sc. Mechatronics, Robotics & Biomechanical Engineering. At Fraunhofer IIS building FIDDLE HMI, the PySide6/ROS 2 operator interface for the HiL-SERL robot-learning pipeline (40 ROS 2 services; 8 crash classes eliminated). M.Eng. research: UAV radar-camera fusion on Jetson Orin NX. Open source: from-scratch HiL-SERL reimplementation in JAX/Flax reaching 55 % task success with zero human input. Seeking working-student roles, continuing alongside the TUM M.Sc. from Oct 2026. Available immediately.

TECHNICAL SKILLS

Robotics: ROS 2, ROS 1 (legacy), Gazebo, MoveIt, SLAM (GMapping, Cartographer, SLAM Toolbox), Sensor Fusion, Radar Odometry, Nav2, PCL, Autonomous Navigation, Motion Planning

Robot Learning & RL: JAX, Flax, SAC, RLPD, HiL-SERL, Behavior Cloning, Reward Classifiers, MuJoCo, Gymnasium

Perception & ML: OpenCV, Azure Kinect, RealSense, 4D Automotive Radar, PyTorch, Computer Vision, Point Cloud Processing

Programming: C++ (proficient, real-time/ROS 2), Python (advanced, tooling & prototyping), PySide6/Qt, MATLAB/Simulink, Bash

DevOps & Tools: Git, Docker, CI/CD (GitLab CI, GitHub Actions), CMake, Linux

Hardware: Ethernet/TCP/UDP, Embedded Systems, PCB Design, 3D Printing

CAD & Simulation: SolidWorks, Onshape, Fusion 360, FEA, PID Control

WORK EXPERIENCE

Robotics Software Intern | Fraunhofer IIS / Adaptive Robotics Research, Dresden

Jun 2026 – present

- Developed FIDDLE HMI, a PySide6 + ROS 2 desktop interface driving the full HiL-SERL robot-learning pipeline on UR cobots, from task configuration and camera setup through reward-classifier training, demo collection, and live inference monitoring.
- Rebuilt the HMI backend from a monolith into a layered transport/clients/facade architecture covering **40 ROS 2 services** (JSON payload transport, typed error hierarchy, cached clients); replaced all direct file access with service calls and shipped it as a standalone ROS 2 package.
- Diagnosed and eliminated **8 Qt/threading SIGSEGV crash classes** (QObject lifetime races between worker threads and widget teardown); centralized the fix into a reusable worker-lifecycle pattern adopted across all dialogs.
- Extended the live ROS 2/MoveIt 2 multi-robot workbench (dual 7-DoF cobots) with runtime planner switching and multi-point Cartesian planning; diagnosed three live robot-run failure modes from logs alone and delivered root-cause analysis to the hardware and perception sub-teams.
- Developed an MCP server exposing robot path planning and task execution as LLM-callable endpoints, enabling natural-language orchestration of the full motion pipeline.

Software Development Intern | TTZ-EMO (Research Institute), THWS

Nov 2024 – Apr 2025

- Ported the GetDP electromagnetic solver to Windows (CMake, MSYS2), resolving POSIX incompatibilities across 3 major library dependencies; produced build documentation that enabled lab-wide cross-platform simulation, eliminating a Linux-only bottleneck for a team of 6 researchers.
- Optimized file I/O via UUID serialization and QThread concurrency in the DST GUI, achieving **35 % faster** load/save throughput across the lab's simulation workflows.
- Developed a static-analysis tool parsing Python/MATLAB codebases (**200+** files) and generating interactive call-graph visualizations (NetworkX + PyVis), giving new team members a navigable map of the codebase. Set up GitLab CI/CD pipeline and Sphinx documentation.

Working Student | TTZ-EMO (Research Institute), THWS

Apr 2024 – Sep 2024

- Integrated multithreaded processing into the Drives Simulation Tool (DST), **accelerating FEM computations by ~40 %**; authored unit tests covering 4 core data-management pipelines and technical documentation with Sphinx/mkdocs.
- Configured a Nextcloud/WebDAV-based Zotero integration, enabling synchronized literature access for a team of 5 researchers and eliminating manual bibliography synchronization.

EDUCATION

M.Sc. Mechatronics, Robotics & Biomechanical Engineering | TUM, Munich

Oct 2026 – Sep 2028 (expected)

M.Eng. Electrical Engineering & IT | THWS

Apr 2026 – Sep 2026

Focus: Drone-Based Multi-Sensor Fusion · transferring to TUM in Oct 2026

B.Eng. Robotics | Technische Hochschule Würzburg-Schweinfurt

Oct 2022 – Mar 2026

Grade: 2.3 · Thesis: 1.3 (Distinction) · 210 ECTS (German scale, 1.0 = best)

Key modules: Localisation & Mapping, Sensor Data Fusion, Control Systems, Image Processing

- Mounted a 4D automotive radar on an Avular Vertex One UAV; built the complete ROS 2 stack converting sparse point clouds into 2D occupancy maps. Engineered 3D-printed sensor mounts and a dual-power interface board supporting both flight and bench operation.
- Fused radar Doppler, IMU, and downward LiDAR for GPS-free dead-reckoning odometry at 50 Hz. Endpoint displacement: **1.9 %**/71 m (room), **3.7 %**/331 m (corridor). Implemented a three-stage multipath-rejection filter achieving **>95 %** spurious-reflection reduction.
- Full pipeline operates at <22 % mean CPU utilization, <100 MB RAM, 50 Hz; all nodes complete within the 20 Hz radar frame budget, confirming feasibility for onboard UAV deployment.

PROJECTS

UAV Radar-Camera Fusion & Multi-Object Tracking (THWS M.Eng. research)

Apr 2026 – Sep 2026

- Built a ROS 2 radar-camera perception stack on an Avular Vertex One UAV (Jetson Orin NX): brought up the Continental ARS548 4D radar over gPTP-synchronized automotive Ethernet, validated PTP time sync live on drone hardware, and designed the sensor mount (CAD).
- Implemented the dockerized calibration pipeline – Kalibr camera intrinsics and trihedral-corner-reflector SVD radar-camera extrinsics with iKalibr cross-check – plus rosbag recording/conversion tooling for calibration and traffic-data runs.
- Began the fusion and tracking layer – YOLO detections associated with radar objects via Hungarian matching, ByteTrack multi-object tracking with radar Doppler in the Kalman update – ahead of the Oct 2026 transfer to TUM.

HIL-SERL Lite (open source)

Jul 2026 – present

- Reimplemented the HIL-SERL human-in-the-loop robot RL pipeline from scratch: SAC in JAX/Flax, RLPD 50/50 demo/online replay sampling, and a CNN vision reward classifier – sized for a laptop and MuJoCo sim.
- Built a YAML task compiler generating multi-stage MuJoCo manipulation scenes around a Franka Panda plus a scripted expert enabling zero-human operation; the fully autonomous demos-to-BC-to-eval pipeline reaches **55 % success with zero human input**, verified by **112+ headless tests**.
- Root-caused a non-monotonic RL training regression on a 2-stage manipulation task (**70 % success collapsing to 0 %** and never recovering, invisible to standard training metrics) via a custom checkpoint-eval probe; fixed with per-stage skill-chaining, restoring **100 % success**.

IDMP (Interactive Distance Field Mapping & Planning) – Grasping, 7-DoF Cobot (Neura MAiRA 7M)

Mar 2025 – Mar 2026

- Migrated IDMP from ROS 1/UR5e (6-DoF) to **ROS 2/MAiRA 7M (7-DoF)**; rebuilt collision pipeline with Azure Kinect + TF2 self-filter, 18 virtual collision points, and null-space exploitation achieving a stable **98–100 Hz** control loop.
- Integrated a Worker Guidance System using MediaPipe hand-gesture recognition to enable hands-free assembly step navigation, paired with a projector overlay for real-time operator feedback.

LANGUAGES

English (C1) | German (B1, actively improving) | Hindi (Native)

INTERESTS

3D Printing · Open-Source Robotics · Embedded Electronics · Cricket & Football · Traveling

Schweinfurt, August 2026



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